

Lesson Plan — Demo Student

Date: 2026-09-08 **Session length:** 60 minutes **Skill(s):** Desmos orientation (no content skill yet — first Desmos exposure)

Objectives

- Get comfortable typing expressions into the SAT's built-in Desmos panel.
- Learn the core moves: graph-and-click for solving, sliders, tables, regression.

Warm-up (5 min)

[Bank empty — no previous skill exists yet for a first lesson, and question-bank/bank.json has no entries either way. Run `/bank <skill>` to populate before a real warm-up can be built. For a genuine Desmos orientation this slot is optional and can also just be "open Desmos and get oriented to the interface" instead of retrieval questions.]

Direct instruction (15 min)

Link: [reference/desmos-playbook.md](#) (this lesson doesn't need a skill notes page — it's the Desmos orientation itself)

- Show the split-screen layout, how to add a line, how to delete a line.
- Universal move: solving any equation by graphing both sides and clicking the intersection.
- Universal move: sliders for unknown constants.

Desmos move demo

Type `y1=2x-3` and `y2=7` on two lines, click the intersection point, read off x . Then add a slider: type `y1=a*x-3`, click "add slider" on `a`, and drag it to see the line rotate.

Guided practice (30 min)

[Bank empty — same as warm-up. Once official questions exist for an early Algebra skill (e.g. Linear equations in one variable), 4–6 of them go here, easy to hard, so the student practices the Desmos moves on real SAT-style problems instead of generic ones.]

In the meantime, a content-free Desmos fluency drill:

1. Graph $y=x^2-4$ and click the two x-intercepts.
2. Graph $y=x^2-4$ and click the vertex.
3. Build a table with x1: 1,2,3,4,5 and y1: 3,5,7,9,11; type $y1 \sim mx1+b$ and read off the slope and intercept.
4. Type $(x-2)^2+(y+1)^2=9$ and identify the center and radius by eye.

Exit ticket (10 min)

1. "Solve $3x + 5 = 2x - 1$ using Desmos. What did you type, and what's x?"
2. "What's one thing Desmos on the SAT can't do that you might expect it to?" (answer: no saving between questions, no image import)

Homework assigned

None this session — first homework goes out after the first content lesson (Linear functions, scheduled 2026-09-10 per schedule.md).

Notes for me after the session

- What clicked / what didn't:
 - Adjust next lesson:
 - Update mastery table? (Y/N)
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Timing check: $5 + 15 + 30 + 10 = 60$ minutes.